Exercise 10

AIM :

To write a C program to perform below

1. add two matrix using pointer
2. multiply two matrix using pointer

ALGORITHM :

Step 1: Define Constants and Matrices

- Define ROWS and COLS as 3 (for 3x3 matrices).
- Initialize two 3x3 integer matrices:
- mat1: {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}
- mat2: {{9, 8, 7}, {6, 5, 4}, {3, 2, 1}}
- Declare a result matrix result of size 3x3 to store operation outputs.

Step 2: Display Input Matrices

- Call displayMatrix for mat1:
- Iterate over each row (i from 0 to 2).
- For each row, iterate over each column (j from 0 to 2).
- Print the element at position (i, j) using pointer arithmetic: *(mat + i * cols + j).
- Print a newline after each row.
- Repeat for mat2.

Step 3: Perform Matrix Addition

- Call addMatrices with pointers to mat1, mat2, result, ROWS, and COLS:
- Iterate over all elements (i from 0 to rows * cols - 1, i.e., 0 to 8).
- For each element, compute: *(result + i) = *(mat1 + i) + *(mat2 + i).
- Display the result matrix using displayMatrix.

Step 4: Perform Matrix Multiplication

- Call multiplyMatrices with pointers to mat1, mat2, result, ROWS, and COLS:
- Iterate over each row of the result (i from 0 to rows-1).

- For each row, iterate over each column of the result (j from 0 to cols-1).
- Initialize *(result + i * cols + j) to 0.
- Perform dot product: Iterate over k from 0 to cols-1, adding (*(mat1 + i * cols + k)) * (*(mat2 + k * cols + j)) to the result element.
- Display the result matrix using displayMatrix.

Step 5: End Program

- Return 0 to indicate successful execution.

OUTPUT :

```
PS D:\C programming\DHAVAMANI_EX10> gcc main.c -o main
PS D:\C programming\DHAVAMANI_EX10> ./main.exe
Matrix 1:
1 2 3
4 5 6
7 8 9

Matrix 2:
9 8 7
6 5 4
3 2 1

Matrix Addition:
10 10 10
10 10 10
10 10 10

Matrix Multiplication:
30 24 18
84 69 54
138 114 90
```